CSA5112 - CRYPTOGRAPHY AND NETWORK SECURITY

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1. Given:

plaintext : MEETATNOON

key : 7

~~Encryption~~

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17

S	T	U	V	W	X	Y	Z
18	19	20	21	22	23	24	25

(i) CAESAR CIPHER:

Encryption : $C = (P+7) \bmod 26$

Decryption : $P = (C-7) \bmod 26$

ENCRYPTION

Plain text	Numeric Value (P)	Formula	Cipher text (C)
M	12	(12+7) mod 26 = 19	T
E	4	(4+7) mod 26 = 11	L
E	4	(4+7) mod 26 = 11	L
T	19	(19+7) mod 26 = 0	A
A	0	(0+7) mod 26 = 7	H
T	19	(19+7) mod 26 = 0	A
N	13	(13+7) mod 26 = 20	U
O	14	(14+7) mod 26 = 21	V
O	14	(14+7) mod 26 = 21	V
N	13	(13+7) mod 26 = 20	U

∴ Ciphe text :
TLLAHAUVVU

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DECRYPTION:

Ciphertext	Numeric value (c)	Formula	Plaintext
T	19	(19-7) mod 26 = 12	M
L	11	(11-7) mod 26 = 4	E
L	11	(11-7) mod 26 = 4	E
A	0	(0-7) mod 26 = 19	T
H	7	(7-7) mod 26 = 0	A
A	0	(0-7) mod 26 = 19	T
U	20	(20-7) mod 26 = 13	N
V	21	(21-7) mod 26 = 14	O
V	21	(21-7) mod 26 = 14	O
U	20	(20-7) mod 26 = 13	N

∴ Plaintext :

MEETATNOON

(ii) VIGENERE CIPHER:

Plaintext : MEETATNOON

key : SUN

	12	4	4	19	0	19	13	14	14	13
Plaintext	M	E	E	T	A	T	N	O	O	N
key	S	U	N	S	U	N	S	U	N	S
	18	20	13	18	20	13	18	20	13	18

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ENCRYPTION:

Plaintext	Numeric (P)	Formula	Ciphertext
M	12	(12+18) mod 26 = 4	E
E	4	(4+20) mod 26 = 24	Y
E	4	(4+13) mod 26 = 17	R
T	19	(19+18) mod 26 = 11	L
A	0	(0+20) mod 26 = 20	U
T	19	(19+13) mod 26 = 6	G
N	13	(13+18) mod 26 = 5	F
O	14	(14+20) mod 26 = 8	I
O	14	(14+13) mod 26 = 1	B
N	13	(13+18) mod 26 = 5	F

∴ Cipher text = EYRLUGFIBF

COMPARISON:

Terms of	Caesar Cipher	Vigenere Cipher
Key space	Only 26 possible keys	key is repeated based on the plaintext.
Repeatation	Same cipher text for repeated letters	Not same cipher text for repeated letters

CONCLUSION:

Cipher text is not suitable because the repeatation of letters as same text. So, there may me chances of decrypt the ~~msg~~ message by attacker. It is not suitable for protecting modern confidential data and its possible key size is only 26.

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2. HILL CIPHER:

GIVEN:

Plain text: MATH

key: $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$

$P_1 = \begin{bmatrix} M \\ A \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \end{bmatrix}$; $P_2 = \begin{bmatrix} T \\ H \end{bmatrix} = \begin{bmatrix} 19 \\ 7 \end{bmatrix}$

$C = PK \bmod 26$ (Encryption)

~~$C_1 = \begin{bmatrix} 12 \\ 0 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} = 36$~~

$C_1 = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 12 \\ 0 \end{bmatrix} = \begin{bmatrix} 36 \\ 60 \end{bmatrix} \bmod 26 = \begin{bmatrix} 10 \\ 8 \end{bmatrix} = \begin{bmatrix} K \\ I \end{bmatrix}$

$C_2 = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 19 \\ 7 \end{bmatrix} = \begin{bmatrix} 71 \\ 144 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 \\ 14 \end{bmatrix} = \begin{bmatrix} T \\ O \end{bmatrix}$

∴ Ciphar text: KITO

Decryption:

$P = K^{-1} C \pmod{26}$

$K^{-1} = \frac{1}{|K|} (adj\ K)$

$|K| = ad - bc = 21 - 10 = 11$

$K^{-1} = \frac{1}{11} \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$

$11^{-1} = 11 \times 19 \bmod 26 = 1$

$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} = \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \bmod 26 = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$

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Cipher text : KITO

Key : $\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$

$C_1 = \begin{bmatrix} K \\ I \end{bmatrix} = \begin{bmatrix} 10 \\ 8 \end{bmatrix}$; $C_2 = \begin{bmatrix} T \\ O \end{bmatrix} = \begin{bmatrix} 19 \\ 14 \end{bmatrix}$

$P_1 = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \end{bmatrix} = \begin{bmatrix} 142 \\ 130 \end{bmatrix} \bmod 26 = \begin{bmatrix} 12 \\ 0 \end{bmatrix} = \begin{bmatrix} M \\ A \end{bmatrix}$

$P_2 = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 19 \\ 14 \end{bmatrix} = \begin{bmatrix} 253 \\ 241 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 \\ 7 \end{bmatrix} = \begin{bmatrix} T \\ H \end{bmatrix}$

∴ The plain text : MATH

If the alternative key, $K = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$

$K^{-1} = \frac{1}{|K|} (adj\ K)$

$|K| = ad - bc = 10 - 12 = -2$

For this there is no gcd with -2. So, the key is unacceptable.

MAJOR WEAKNESS OF HILL CIPHER :

* It is linearity
* Easy to decrypt the message for the attacker

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3.

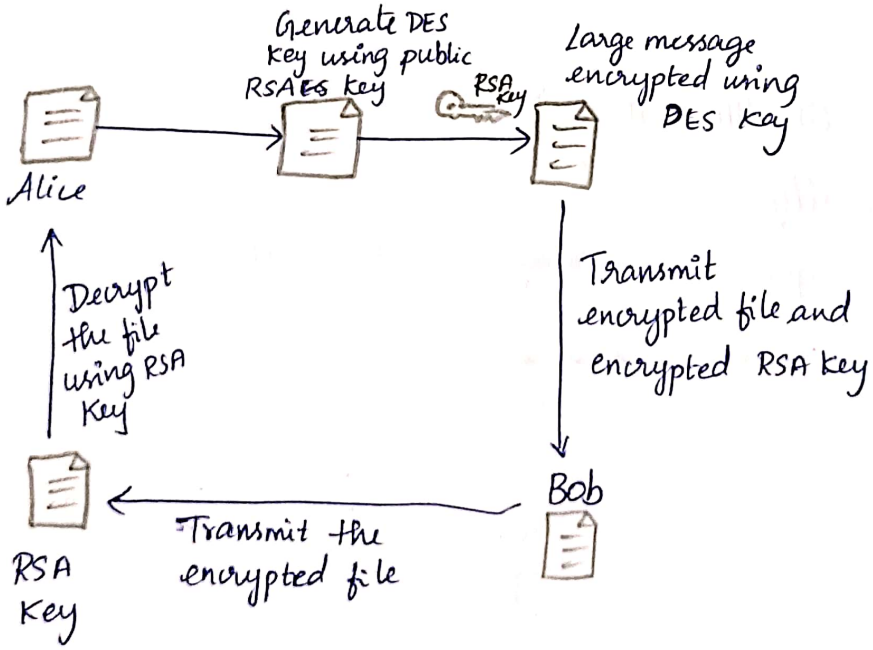
(a)

AES - 256 - GCM	RSA
*. It is symmetric key encryption algorithm	*. It is Asymmetric key encryption algorithm
*. It uses only one key for both encryption and decryption process.	*. It uses two keys as private and public
*. It is relatively quicker	*. It is a slow algorithm which increases the encryption cost.
*. Suitable for layer MRI fees.	*. Not suitable for layer MRI fees.

No. of keys Required:

AES: $\frac{n(n-1)}{2} = \frac{200(199)}{2} = 19900$ keys

(b). HYBRID ENCRYPTION WORKFLOW:



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4. RSA ALGORITHM:

Given:

$p = 11$

$q = 19$

$e = 7$

Compute n.

$n = p \times q$

$\boxed{n = 209}$

Compute $\phi(n)$.

$\phi(n) = (p-1)(q-1) = 10 \times 18 = 180$

Choose e

$1 < e < \phi(n)$ → valid

$\gcd(e, \phi(n)) = 1$

$\gcd(7, 180) = 1$ → valid

Compute d.

$e \cdot d \equiv 1 \pmod{\phi(n)}$

$7 \times d \equiv 1 \pmod{180}$

$\boxed{d = 103}$

Public key: $(e, n) = (7, 209)$

Private key: $(d, n) = (103, 209)$

ENCRYPTION:

$M = 5$

$C = M^e \bmod n = 5^7 \bmod 209 = 78125 \bmod 209 = 168$

DECRYPTION:

$M = C^d \bmod n = 168^{103} \bmod 209 = 5$

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5.

(i) CAESAR CIPHER :

Plaintext : HELLO

Key : 3.

Encryption : $C = (P + 3) \bmod 26$

Plain text	Numeric value (P)	Formula	Cipher text (C)
H	7	(7+3) mod 26 = 10	K
E	4	(4+3) mod 26 = 7	H
L	11	(11+3) mod 26 = 14	O
L	11	(11+3) mod 26 = 14	O
O	14	(14+3) mod 26 = 17	R

∴ Cipher text : KHOOR

Decryption : $P = (C - 3) \bmod 26$

Plain text	Numeric value (C)	Formula	Plain text (P)
K	10	(10-3) mod 26 = 7	H
H	7	(7-3) mod 26 = 4	E
O	14	(14-3) mod 26 = 11	L
O	14	(14-3) mod 26 = 11	L
R	17	(17-3) mod 26 = 14	O

∴ Plain text : HELLO

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(ii) PLAYFAIR CIPHER:

let, plaintext : JAVASCRIPT

Key : INVISIBLE

Keyword :

INVSBLE

I	N	V	S	B
L	E	A	C	D
F	G	H	K	M
O	P	Q	R	T
U	W	X	Y	Z

Plaintext : JA|VA|SC|RI|PT → IA|VA|SC|RI|PT

Pair 1 : IA

Using Rectangular Rule;

I → V

A → L

Pair 2 : VA

Using Column Rule;

V → A

A → H

Pair 3 : SC

Using Column Rule;

S → C

C → K

Pair 4 : RI

Using Rectangle Rule;

R → O

I → S

Pair 5 : PT

Using Row Rule;

P → Q

T → O

∴ Cipher text : VLAHCKOSQO

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(iii) VIGENERE CIPHER:

let, Plaintext : MEETATNOON

Key : CIPHER

	12	4	4	19	0	19	13	14	14	13
Plaintext	M	E	E	T	A	T	N	O	O	N
key	C	I	P	H	E	R	C	I	P	H
	2	8	15	7	4	17	2	8	15	7

Encryption: $C = (P + K) \bmod 26$

Plain text	Numeric (P)	Formula	Cipher text
M	12	$(12+2) \bmod 26 = 14$	O
E	4	$(4+8) \bmod 26 = 12$	M
E	4	$(4+15) \bmod 26 = 19$	T
T	19	$(19+7) \bmod 26 = 0$	A
A	0	$(0+4) \bmod 26 = 4$	E
T	19	$(19+17) \bmod 26 = 10$	K
N	13	$(13+2) \bmod 26 = 15$	P
O	14	$(14+8) \bmod 26 = 22$	W
O	14	$(14+15) \bmod 26 = 3$	D
N	13	$(13+7) \bmod 26 = 20$	U

∴ The cipher text: OMTAEKPWDU

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